

Enterprise Detection Engineering & Threat Hunting Lab

Executive Summary

Purpose

The **Enterprise Detection Engineering & Threat Hunting Lab** was developed to simulate a modern Security Operations Center (SOC) environment focused on endpoint visibility, detection engineering, and proactive threat hunting. The project demonstrates how security telemetry can be collected, analyzed, and transformed into actionable detections using industry-standard tools and methodologies.

Overview

The laboratory uses **Microsoft Windows** as the monitored endpoint, **Sysmon** for enhanced endpoint telemetry, **Splunk Universal Forwarder** for centralized log collection, and **Splunk Enterprise** as the Security Information and Event Management (SIEM) platform. Security events are collected, indexed, and analyzed using Splunk Search Processing Language (SPL) to identify suspicious activities, validate detections, and support threat hunting investigations.

Project Objectives

- Build a centralized endpoint monitoring solution.
- Collect high-fidelity Windows security telemetry.
- Develop behavioral detection rules using SPL.
- Perform proactive threat hunting based on endpoint activity.
- Map detections to the MITRE ATT&CK framework.
- Create dashboards that improve SOC visibility.
- Document the implementation using enterprise documentation standards.

Core Components

Component	Purpose
Microsoft Windows	Generates endpoint security events

Sysmon	Provides detailed endpoint telemetry
Splunk Universal Forwarder	Collects and forwards logs
Splunk Enterprise	Centralized SIEM and analytics platform
SPL	Detection engineering and threat hunting
MITRE ATT&CK	Detection mapping and coverage analysis

Key Capabilities

- Centralized log collection
- Endpoint telemetry monitoring
- Behavioral detection engineering
- Threat hunting with SPL
- Dashboard visualization
- Security event investigation
- MITRE ATT&CK mapping
- Detection validation through attack simulation

Business Value

The project demonstrates practical SOC workflows used in enterprise environments, including telemetry collection, security monitoring, behavioral detection, investigation, and continuous improvement of detection capabilities. It provides hands-on experience with technologies and processes commonly used by Security Operations Centers to identify and investigate malicious activity.

Outcome

The completed laboratory provides a functional security monitoring environment capable of collecting endpoint telemetry, detecting suspicious behavior, supporting proactive threat hunting, and documenting detection logic using professional SOC practices. The project serves as a practical demonstration of detection engineering principles and enterprise security operations.

Solution Architecture

Purpose

The laboratory architecture is designed to simulate the core workflow of an enterprise Security Operations Center (SOC). It provides a centralized pipeline for collecting endpoint telemetry, processing security events, developing detections, and performing threat hunting using Splunk Enterprise.

Architecture Overview

The environment follows a centralized monitoring model where endpoint telemetry is generated, securely forwarded, indexed, and analyzed within the SIEM platform. Each component performs a dedicated role to ensure reliable log collection, efficient event analysis, and accurate security monitoring.

Architecture Components

Component	Function
Microsoft Windows Endpoint	Generates operating system and security events
Sysmon	Captures detailed endpoint telemetry including process creation, network connections, registry modifications, and file activities
Splunk Universal Forwarder	Collects Windows and Sysmon logs and securely forwards them to Splunk Enterprise
Splunk Enterprise	Indexes, stores, searches, and analyzes collected telemetry
Detection Rules (SPL)	Detect suspicious behaviors using behavioral analytics
Dashboards	Visualize security events and monitoring metrics

Threat Hunting
Queries

Identify hidden threats through proactive investigations

Data Flow

Windows Endpoint

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Sysmon

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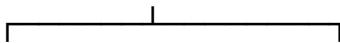
Splunk Universal Forwarder

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Splunk Enterprise

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Detection Rules Dashboards

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Threat Hunting & Investigation

Security Workflow

1. Endpoint activity generates security events.
2. Sysmon records detailed telemetry.
3. Splunk Universal Forwarder forwards logs to Splunk Enterprise.
4. Splunk indexes and normalizes incoming data.

5. SPL queries analyze telemetry for suspicious behavior.
6. Detection rules generate security alerts.
7. Analysts investigate events through dashboards and threat hunting queries.

Security Benefits

- Centralized security monitoring
- Enhanced endpoint visibility
- High-fidelity telemetry collection
- Faster detection of suspicious activities
- Efficient security investigations
- Scalable detection engineering workflow
- Improved SOC operational visibility

Architecture Summary

This architecture establishes a complete telemetry pipeline that transforms raw endpoint events into actionable security intelligence. The modular design reflects enterprise SOC practices and provides a scalable foundation for detection engineering, threat hunting, and incident investigation.

Lab Environment

Purpose

The laboratory environment provides a controlled platform for implementing detection engineering, validating security detections, and performing threat hunting without impacting production systems.

Environment Overview

The lab consists of a Microsoft Windows endpoint configured with Sysmon for enhanced telemetry collection and Splunk Enterprise for centralized log analysis. Security events generated by the endpoint are forwarded through Splunk Universal Forwarder, enabling near real-time monitoring and investigation.

Environment Components

Component	Description
Operating System	Microsoft Windows
Endpoint Monitoring	Sysmon
SIEM Platform	Splunk Enterprise
Log Collector	Splunk Universal Forwarder
Query Language	Splunk Search Processing Language (SPL)
Detection Framework	MITRE ATT&CK

Lab Objectives

- Collect high-fidelity endpoint telemetry.
- Centralize Windows and Sysmon event logs.
- Develop and validate behavioral detections.
- Perform proactive threat hunting.
- Build operational dashboards for SOC monitoring.

Monitoring Scope

The laboratory monitors security-relevant endpoint activities, including:

- Process creation
- Command-line execution
- Parent-child process relationships
- Network connections
- Registry modifications
- File creation events
- Privileged process execution
- PowerShell activity
- Windows Security Events
- Sysmon Operational Events

Event Collection Workflow

Windows Endpoint

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Windows Logs + Sysmon Logs

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Splunk Universal Forwarder

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Splunk Enterprise Index

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Search • Detection • Dashboard • Investigation

Expected Outcomes

The laboratory provides a realistic environment for collecting endpoint telemetry, validating detection rules, investigating suspicious activities, and improving security visibility through centralized monitoring.

Implementation Benefits

- Safe environment for security testing
- Consistent telemetry collection
- Centralized log management
- Efficient detection validation
- Repeatable threat hunting workflow
- Practical SOC experience

Sysmon Deployment

Purpose

Sysmon (System Monitor) was deployed to enhance endpoint visibility by collecting detailed system activity that is not available through standard Windows event logging. It serves as the primary telemetry source for detection engineering, threat hunting, and incident investigations.

Overview

Sysmon continuously monitors endpoint activity and records high-value security events in the Windows Event Log. These events provide detailed context about system behavior, enabling analysts to detect suspicious activities and investigate security incidents with greater accuracy.

Deployment Configuration

Configuration	Details
Deployment Type	Windows System Service
Log Source	Microsoft-Windows-Sysmon/Operational
Event Collection	Continuous
Log Destination	Splunk Enterprise
Collection Method	Splunk Universal Forwarder

Monitored Activities

The Sysmon configuration collects telemetry related to:

- Process creation
- Process termination

- Network connections
- Driver loading
- Image loading
- File creation
- Registry modifications
- Process access events
- DNS queries (if configured)
- Parent-child process relationships
- Command-line execution

Operational Workflow

System Activity

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Sysmon

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Generates Event Logs

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Splunk Universal Forwarder

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Splunk Enterprise

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Detection Rules & Threat Hunting

Security Benefits

- Provides high-fidelity endpoint telemetry.
- Improves visibility into attacker behavior.
- Supports behavioral detection engineering.

- Enables detailed forensic investigations.
- Reduces dependency on default Windows logging.
- Strengthens threat hunting capabilities.

Implementation Summary

Sysmon forms the foundation of the laboratory's monitoring strategy by generating detailed endpoint telemetry required for behavioral detections and proactive threat hunting. Its integration with Splunk Enterprise enables efficient collection, analysis, and investigation of security events within a centralized SOC environment.

Splunk Universal Forwarder Configuration

Purpose

Splunk Universal Forwarder (UF) is responsible for collecting Windows and Sysmon event logs from the monitored endpoint and securely forwarding them to Splunk Enterprise for centralized analysis.

Overview

The Universal Forwarder is a lightweight data collection agent designed to continuously monitor configured log sources with minimal impact on system performance. It enables reliable and near real-time transmission of security telemetry to the SIEM platform.

Configuration Details

Configuration	Details
Forwarder	Splunk Universal Forwarder
Source Logs	Windows Event Logs, Sysmon Operational Logs
Destination	Splunk Enterprise
Collection Mode	Continuous
Data Transfer	Near Real-Time

Collected Log Sources

The following log sources are configured for forwarding:

- Windows Security Event Log

- Windows System Event Log
- Windows Application Event Log
- Sysmon Operational Event Log

Data Collection Workflow

Windows Event Logs

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Sysmon Operational Logs

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Splunk Universal Forwarder

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Secure Log Forwarding

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Splunk Enterprise

Operational Benefits

- Continuous log collection
- Centralized event forwarding
- Lightweight endpoint agent
- Reliable telemetry delivery
- Low resource utilization
- Simplified log management

Security Benefits

- Ensures security events are available for analysis in near real time.
- Reduces the risk of missing critical endpoint activity.
- Supports centralized monitoring across multiple log sources.

- Provides a reliable telemetry pipeline for detection engineering and threat hunting.

Implementation Summary

Splunk Universal Forwarder serves as the communication layer between the monitored endpoint and Splunk Enterprise. By continuously forwarding Windows and Sysmon logs, it establishes a dependable telemetry pipeline that supports detection development, dashboard visualization, security investigations, and proactive threat hunting.

Splunk Enterprise Configuration

Purpose

Splunk Enterprise serves as the centralized Security Information and Event Management (SIEM) platform responsible for ingesting, indexing, analyzing, and visualizing security telemetry collected from monitored endpoints.

Overview

Security events generated by Windows and Sysmon are forwarded to Splunk Enterprise, where they are indexed into searchable repositories. Analysts use Splunk Search Processing Language (SPL) to investigate events, develop detection rules, build dashboards, and perform proactive threat hunting.

Platform Functions

Function	Description
Data Ingestion	Receives logs from Splunk Universal Forwarder
Data Indexing	Stores and organizes security events
Search & Analysis	Executes SPL queries for investigations
Detection Engineering	Develops behavioral detection rules

Dashboarding Visualizes security metrics and events

Threat Hunting Supports proactive security investigations

Data Processing Workflow

Log Collection

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Data Ingestion

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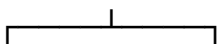
Event Indexing

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SPL Search & Analysis

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Detections Dashboards

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Security Investigation

Core Capabilities

- Centralized log management
- High-speed event searching

- SPL-based analytics
- Behavioral detection development
- Interactive dashboards
- Event correlation
- Threat hunting
- Incident investigation

Operational Benefits

- Consolidates security telemetry into a single platform.
- Accelerates event analysis and investigations.
- Simplifies detection development using SPL.
- Improves analyst visibility across endpoint activities.
- Enables rapid identification of suspicious behavior.

Key Takeaways

- Splunk Enterprise acts as the central analytics engine for the laboratory.
- Indexed telemetry enables efficient searching, correlation, and investigation.
- SPL provides the flexibility required to build custom detections and threat hunting queries.
- Centralized monitoring improves security visibility and supports faster incident analysis.

Detection Engineering Methodology

Purpose

The objective of the detection engineering process is to develop reliable, behavior-based detections that identify suspicious activities using endpoint telemetry collected from Sysmon and Windows Event Logs.

Methodology Overview

Each detection was developed using a structured workflow to ensure accuracy, consistency, and operational value. Rather than relying solely on predefined signatures, detections focus on identifying attacker behaviors that are commonly observed during real-world security incidents.

Detection Development Lifecycle

Threat Scenario

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Identify Telemetry Source

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Analyze Relevant Events

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Develop SPL Query

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Validate Detection

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Reduce False Positives

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Deploy Detection

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Continuous Improvement

Detection Workflow

Phase	Description
Threat Identification	Select an attacker technique or suspicious behavior to monitor.
Telemetry Analysis	Identify the Sysmon or Windows Event IDs required for detection.
Query Development	Create an SPL query to detect the target behavior.
Validation	Execute controlled attack simulations to verify detection accuracy.
Tuning	Refine the query to minimize false positives while maintaining detection coverage.
Monitoring	Continuously review detection performance and improve detection logic when necessary.

Detection Design Principles

The detection rules developed within this laboratory follow the following principles:

- Behavior-based detection over static signatures.
- High-confidence telemetry sources.
- Clear and maintainable SPL queries.
- Low false-positive rate.
- Easy investigation and validation.
- Alignment with enterprise SOC monitoring practices.

Expected Outcomes

The methodology enables security analysts to:

- Detect suspicious endpoint activity.
- Improve investigation efficiency.
- Validate detections using realistic attack simulations.
- Build reusable and maintainable detection logic.
- Continuously enhance detection coverage.

Key Takeaways

- Effective detections begin with high-quality telemetry.
- Detection rules should focus on attacker behavior rather than isolated events.
- Validation and tuning are essential before deploying a detection.
- Detection engineering is an ongoing process that requires continuous improvement.

Threat Hunting Methodology

Purpose

The threat hunting process is designed to proactively identify suspicious or malicious activities that may not trigger predefined detection rules. It complements automated detections by enabling analysts to investigate abnormal behaviors using endpoint telemetry and analytical queries.

Methodology Overview

Threat hunting within this laboratory follows a hypothesis-driven approach. Instead of waiting for security alerts, analysts actively search for indicators of compromise by examining Windows and Sysmon telemetry, correlating security events, and validating suspicious activities through structured investigations.

Threat Hunting Workflow

Define Hunting Hypothesis

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Identify Data Sources

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Develop SPL Query

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Analyze Results

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Validate Findings

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Hunting Process

Phase	Description
Hypothesis	Define a suspicious behavior or attack technique to investigate.
Data Collection	Identify relevant Windows and Sysmon events.
Query Development	Create SPL queries to search for suspicious activity.
Analysis	Review results, identify anomalies, and correlate related events.
Validation	Verify findings through additional telemetry and context.
Documentation	Record observations, conclusions, and recommended improvements.

Hunting Scenarios

The laboratory includes proactive hunting for:

- Suspicious PowerShell execution
- Encoded command execution
- Unusual parent-child process relationships
- Unexpected network connections
- Privileged process execution
- Persistence-related activities
- Command-line anomalies
- Reconnaissance behavior

Benefits

- Detects activity that may bypass existing detection rules.
- Improves analyst understanding of endpoint behavior.
- Identifies detection gaps and opportunities for improvement.
- Supports continuous enhancement of detection coverage.
- Strengthens overall SOC visibility and investigation capability.

Key Takeaways

- Threat hunting is a proactive security activity rather than an alert-driven process.
- A hypothesis-driven approach improves investigation quality and consistency.
- Effective hunting depends on high-quality endpoint telemetry and well-designed analytical queries.
- Findings from threat hunts can be used to create new detection rules and improve existing detections.

Detection 01 – Suspicious PowerShell Execution

Objective

Detect PowerShell processes that may indicate malicious or unauthorized activity by monitoring command-line execution and process creation events.

Threat Description

PowerShell is a legitimate Windows administration tool frequently abused by attackers for reconnaissance, payload execution, privilege escalation, persistence, and defense evasion. Monitoring PowerShell activity helps identify suspicious behavior at an early stage of an attack.

Data Source

Source	Value
Telemetry	Sysmon
Log Source	Microsoft-Windows-Sysmon/Operational
Event ID	1 (Process Creation)

MITRE ATT&CK Mapping

Tactic	Technique	ID
Execution	PowerShell	T1059.001

Detection Logic

The detection identifies PowerShell process creation events and captures important context such as the executing user, host, parent process, executable path, and command-line arguments. This information helps analysts determine whether the execution is legitimate or requires further investigation.

SPL Query

```
index=main sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"  
EventCode=1 Image="*\\powershell.exe"
```

```
| table _time Computer User ParentImage Image CommandLine
```

Investigation Steps

1. Verify the user account that executed PowerShell.
2. Review the parent process that launched PowerShell.
3. Examine the full command-line arguments.
4. Determine whether the activity matches normal administrative behavior.
5. Investigate any encoded commands, download activity, or unusual execution patterns.

Expected Result

The query returns PowerShell execution events with sufficient context to support investigation and identify potentially suspicious activity.

Analyst Notes

PowerShell execution is not inherently malicious. Detection results should always be evaluated in the context of the user, host, command-line arguments, parent process, and expected administrative activity before determining whether an incident has occurred.

Key Takeaways

- PowerShell is a high-value telemetry source for endpoint monitoring.
- Command-line visibility significantly improves investigation quality.
- Parent-child process relationships provide valuable execution context.
- Behavioral analysis reduces unnecessary false positives.

Detection 02 – Encoded PowerShell Command Detection

Objective

Identify PowerShell executions that use Base64-encoded commands, a technique commonly employed by attackers to conceal malicious activity and evade basic detection mechanisms.

Threat Description

Attackers frequently execute PowerShell with the **-EncodedCommand** parameter to hide scripts and bypass simple command-line monitoring. Detecting encoded commands helps identify suspicious execution patterns that may indicate malware, post-exploitation activity, or unauthorized administrative actions.

Data Source

Source		Value
Telemetry	Sysmon	
	Log Source	Microsoft-Windows-Sysmon/Operational
Event ID	1 (Process Creation)	

MITRE ATT&CK Mapping

Tactic	Technique	ID
Defense Evasion	PowerShell	T1059.001

Detection Logic

The detection monitors PowerShell process creation events and searches the command line for indicators such as **-EncodedCommand**, **-enc**, or similar encoded execution parameters. Any matching event is flagged for analyst review.

SPL Query

```
index=main sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"  
EventCode=1
```

```
(Image="*\\powershell.exe" OR OriginalFileName="PowerShell.EXE")
```

```
(CommandLine="*-EncodedCommand*" OR CommandLine="*-enc*")
```

```
| table _time Computer User ParentImage Image CommandLine
```

Investigation Steps

1. Review the complete PowerShell command line.
2. Identify the parent process that launched PowerShell.
3. Verify whether the execution originated from an authorized administrator.
4. Decode the Base64 command (if appropriate) to determine its intended behavior.
5. Correlate the event with additional Sysmon or Windows Security events to assess the overall activity.

Expected Result

The detection identifies PowerShell executions using encoded commands and provides sufficient context for analysts to determine whether the activity is legitimate or potentially malicious.

Analyst Notes

Encoded PowerShell commands are not always malicious. Some enterprise management tools and automation platforms legitimately use encoded commands. Analysts should validate the execution context before classifying the event as a security incident.

Key Takeaways

- Encoded commands are a common technique used to obscure malicious activity.
- Command-line analysis provides valuable context during investigations.
- Correlating parent process and user information improves detection accuracy.
- Validation helps reduce false positives while maintaining strong detection coverage.

Detection 03 – LSASS Memory Dump Detection

Objective

Detect attempts to access or dump the Local Security Authority Subsystem Service (LSASS) process, which may indicate credential theft activity.

Threat Description

The LSASS process stores authentication-related information in memory. Attackers often target LSASS to extract user credentials, NTLM hashes, or Kerberos tickets for privilege escalation and lateral movement. Monitoring access to this process helps identify potential credential dumping attempts.

Data Source

Source	Value
Telemetry	Sysmon
Log Source	Microsoft-Windows-Sysmon/Operational
Event ID	10 (Process Access)

MITRE ATT&CK Mapping

Tactic	Technique	ID
Credential Access	OS Credential Dumping	T1003.001

Detection Logic

The detection monitors Sysmon Process Access events where a process attempts to access **lsass.exe**. Events are reviewed to identify unauthorized processes requesting access rights commonly associated with credential dumping.

SPL Query

```
index=main sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"  
EventCode=10 TargetImage="*\\lsass.exe"
```

```
| table _time Computer SourceImage TargetImage GrantedAccess User
```

Investigation Steps

1. Identify the process requesting access to **lsass.exe**.
2. Verify whether the process is a trusted security or administration tool.
3. Review the user account associated with the process.
4. Check the GrantedAccess value to determine the level of access requested.
5. Correlate the event with other endpoint activities such as PowerShell execution, suspicious process creation, or network connections.

Expected Result

The query identifies processes attempting to access LSASS memory, allowing analysts to distinguish legitimate administrative activity from potential credential dumping attempts.

Analyst Notes

Not every LSASS access event is malicious. Security software, backup solutions, and authorized administration tools may legitimately access the process. Investigation should always consider process reputation, execution context, and related endpoint activity before determining whether an incident has occurred.

Key Takeaways

- LSASS is a high-value target for attackers seeking credentials.
- Monitoring Process Access events improves visibility into credential theft attempts.
- Event correlation is essential to accurately assess suspicious LSASS access.
- High-confidence detections reduce false positives while improving investigation efficiency.

Detection 04 – Registry Persistence

Detection

Objective

Detect modifications to Windows Registry Run and RunOnce keys that may indicate an attempt to establish persistence on the endpoint.

Threat Description

Attackers frequently create or modify Registry Run keys to automatically execute malicious programs whenever a user logs on or the system starts. Monitoring these registry locations helps identify unauthorized persistence mechanisms before they can be used to maintain long-term access.

Data Source

Source	Value
Telemetry	Sysmon
Log Source	Microsoft-Windows-Sysmon/Operational
Event ID	13 (Registry Value Set)

MITRE ATT&CK Mapping

Tactic	Technique	ID
Persistence	Registry Run Keys / Startup Folder	T1547.001

Detection Logic

The detection monitors registry value modifications targeting Windows **Run** and **RunOnce** locations. When a new value is created or an existing value is modified, the event is captured with relevant context, including the affected registry path, process, and user.

SPL Query

```
index=main sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"  
EventCode=13
```

```
(TargetObject="*\\Software\\Microsoft\\Windows\\CurrentVersion\\Run*" OR
```

```
TargetObject="*\\Software\\Microsoft\\Windows\\CurrentVersion\\RunOnce*")
```

```
| table _time Computer User Image TargetObject Details
```

Investigation Steps

1. Review the modified registry path.
2. Identify the process responsible for the change.
3. Verify the user account that performed the action.
4. Examine the executable or script referenced in the registry value.
5. Determine whether the persistence mechanism is authorized or suspicious.

Expected Result

The query identifies registry modifications that may establish automatic program execution during user logon or system startup, enabling analysts to investigate potential persistence activity.

Analyst Notes

Registry Run keys are also used by legitimate applications during installation and software updates. Analysts should validate the publisher, executable path, digital signature, and business purpose before classifying the event as malicious.

Key Takeaways

- Registry Run keys are a common persistence mechanism.
- Sysmon Event ID 13 provides valuable visibility into registry modifications.
- Combining registry events with process creation telemetry improves investigation accuracy.
- Validation and context are essential to reduce false positives.

Detection 05 – Suspicious System Discovery Commands

Objective

Detect the execution of common Windows reconnaissance commands that may indicate system or network discovery activity.

Threat Description

After gaining access to a system, attackers typically perform reconnaissance to understand the environment before moving laterally or escalating privileges. Commands such as **whoami**, **hostname**, **ipconfig**, **net user**, **net localgroup**, and **systeminfo** are frequently used to collect information about users, network configuration, and system details.

Data Source

Source	Value
Telemetry	Sysmon
Log Source	Microsoft-Windows-Sysmon/Operational
Event ID	1 (Process Creation)

MITRE ATT&CK Mapping

Tactic	Technique	ID
Discovery	System Owner/User Discovery	T1033
Discovery	System Information Discovery	T1082

Detection Logic

The detection monitors process creation events and identifies the execution of common reconnaissance utilities. It captures the command line, parent process, user account, and host information to provide analysts with sufficient context for investigation.

SPL Query

```
index=main sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"
EventCode=1
```

```
(Image="*\whoami.exe" OR Image="*\hostname.exe" OR Image="*\ipconfig.exe" OR
Image="*\systeminfo.exe" OR Image="*\net.exe")
```

```
| table _time Computer User ParentImage Image CommandLine
```

Investigation Steps

1. Identify the executed discovery command.
2. Review the user account responsible for the execution.
3. Examine the parent process that launched the command.
4. Determine whether the activity matches normal administrative operations.
5. Correlate the event with other suspicious behaviors such as PowerShell execution, registry persistence, or unusual network activity.

Expected Result

The query identifies endpoint reconnaissance commands and provides the necessary context to determine whether the activity is part of legitimate administration or a potential attack.

Analyst Notes

Administrative users and IT management tools may legitimately execute these commands. Investigation should focus on execution frequency, parent process, user context, and correlation with other suspicious events rather than the command alone.

Key Takeaways

- Reconnaissance is often one of the earliest stages of an attack.
- Process creation telemetry provides strong visibility into discovery activities.
- Context and event correlation are essential for accurate analysis.
- Monitoring common discovery commands helps identify attacker behavior before later attack stages.

Detection 07 – Failed Logon Detection

Objective

Detect repeated Windows authentication failures that may indicate brute-force attacks, password spraying, or unauthorized access attempts.

Threat Description

Failed authentication attempts are common indicators of malicious activity. Attackers often attempt multiple logins using different passwords or user accounts to gain unauthorized access. Monitoring failed logons enables early detection of these attack techniques before a successful compromise occurs.

Data Source

Source	Value
Telemetry	Windows Security Event Log
Log Source	Security
Event ID	4625 (An account failed to log on)

MITRE ATT&CK Mapping

Tactic	Technique	ID
Credential Access	Brute Force	T1110

Detection Logic

The detection monitors Windows Security Event ID **4625** and identifies accounts or hosts generating multiple failed authentication attempts within a defined time period. Excessive failures may indicate brute-force activity or invalid authentication attempts requiring investigation.

SPL Query

```
index=main sourcetype="WinEventLog:Security" EventCode=4625
```

```
| stats count by Account_Name, Source_Network_Address, host
```

```
| where count >= 5
```

```
| sort - count
```

Investigation Steps

1. Identify the affected user account.
2. Review the source IP address or originating host.
3. Determine whether multiple accounts were targeted.
4. Check for any successful logon events (Event ID 4624) following the failures.
5. Investigate related endpoint activity such as PowerShell execution, privileged logons, or suspicious process creation.

Expected Result

The query identifies accounts or systems experiencing repeated authentication failures, allowing analysts to quickly identify potential brute-force or password spraying activity.

Analyst Notes

Failed logons are not always malicious. Users may mistype passwords, have expired credentials, or experience application authentication issues. Analysts should consider the number of failures, source system, affected accounts, and subsequent authentication activity before escalating the event.

Key Takeaways

- Event ID 4625 is a critical authentication event for SOC monitoring.
- Multiple failed logons from the same source may indicate brute-force activity.
- Correlation with successful logons and endpoint telemetry improves investigation accuracy.

- Authentication monitoring is a foundational capability of enterprise Security Operations Centers.

Detection 08 – Suspicious Network Connection Detection

Objective

Detect outbound network connections initiated by monitored processes to improve visibility into potential command-and-control (C2) communication or unauthorized network activity.

Threat Description

After compromising a system, attackers often establish outbound network connections to communicate with remote infrastructure, download additional payloads, or exfiltrate data. Monitoring process-initiated network connections helps identify abnormal communication patterns that may require investigation.

Data Source

Source	Value
Telemetry	Sysmon
Log Source	Microsoft-Windows-Sysmon/Operational
Event ID	3 (Network Connection)

MITRE ATT&CK Mapping

Tactic	Technique	ID
Command and Control	Application Layer Protocol	T1071

Detection Logic

The detection monitors Sysmon Network Connection events and records the initiating process, destination IP address, destination port, and user context. These events provide visibility into outbound communications initiated by endpoint processes.

SPL Query

```
index=main sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"  
EventCode=3  
| table _time Computer User Image DestinationIp DestinationPort Protocol
```

Investigation Steps

1. Identify the process initiating the network connection.
2. Review the destination IP address and destination port.
3. Verify whether the destination is expected within the environment.
4. Correlate the event with recent process creation and PowerShell activity.
5. Investigate repeated or unusual outbound connections.

Expected Result

The query displays outbound network connections with sufficient process context to support threat hunting and incident investigations.

Analyst Notes

Not all outbound network connections are suspicious. Browsers, update services, and enterprise applications routinely establish external connections. Investigation should focus on unusual destinations, unexpected processes, or communication patterns that deviate from the established baseline.

Key Takeaways

- Network telemetry provides valuable visibility into endpoint communications.
- Correlating network events with process creation improves investigation accuracy.
- Continuous monitoring of outbound connections supports early identification of command-and-control activity.
- Behavioral analysis helps distinguish legitimate traffic from suspicious communications.

Detection 09 – Account Lockout Detection

Objective

Detect Windows account lockout events that may indicate brute-force attacks, password spraying, or repeated authentication failures.

Threat Description

Account lockouts occur when multiple failed authentication attempts exceed the configured security threshold. While lockouts may result from user error, they can also indicate malicious attempts to gain unauthorized access using automated password guessing techniques.

Data Source

Source	Value
Telemetry	Windows Security Event Log
Log Source	Security
Event ID	4740 (A user account was locked out)

MITRE ATT&CK Mapping

Tactic	Technique	ID
Credential Access	Brute Force	T1110

Detection Logic

The detection monitors Windows Security Event ID **4740** and records account lockout events, including the affected user account, source computer, and timestamp. These events are reviewed to identify abnormal authentication activity and support investigation of potential brute-force attacks.

SPL Query

```
index=main sourcetype="WinEventLog:Security" EventCode=4740
```

| table _time host TargetUserName CallerComputerName

Investigation Steps

1. Identify the locked user account.
2. Review the originating computer or system.
3. Correlate the lockout with recent failed logon events (Event ID 4625).
4. Determine whether multiple accounts experienced lockouts.
5. Investigate any successful authentication events following the lockout.

Expected Result

The query identifies Windows account lockout events, allowing analysts to quickly investigate suspicious authentication activity and determine whether the lockout resulted from user error or malicious behavior.

Analyst Notes

Account lockouts are not always indicators of compromise. Expired credentials, cached passwords, scheduled tasks, or user mistakes may generate legitimate lockout events. Analysts should always correlate lockouts with additional authentication events before escalating an incident.

Key Takeaways

- Account lockouts provide early visibility into authentication-related attacks.
- Event correlation improves the accuracy of security investigations.
- Monitoring Event ID 4740 strengthens identity and access monitoring.
- Authentication telemetry is a critical component of enterprise SOC operations.

Detection 10 – Living Off the Land Binary (LOLBin) Execution Detection

Objective

Detect the execution of commonly abused Windows native binaries (LOLBins) that attackers use to execute malicious code, evade security controls, or perform post-exploitation activities.

Threat Description

Living Off the Land Binaries (LOLBins) are legitimate Windows executables that can be misused by attackers to perform malicious actions without introducing external malware. Because these binaries are trusted by the operating system, they are frequently used to evade security controls and blend into normal system activity.

Common examples include:

- powershell.exe
- cmd.exe
- rundll32.exe
- regsvr32.exe
- mshta.exe
- certutil.exe
- wmic.exe
- bitsadmin.exe

Monitoring the execution of these binaries improves visibility into attacker techniques while supporting behavioral detection and threat hunting.

Data Source

Source		Value
Telemetry	Sysmon	
Log Source	Microsoft-Windows-Sysmon/Operational	
Event ID	1 (Process Creation)	

MITRE ATT&CK Mapping

Tactic	Technique	ID
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Defense Evasion	System Binary Proxy Execution	T1218
Execution	Command and Scripting Interpreter	T1059
Command and Control	Ingress Tool Transfer	T1105

Detection Logic

The detection monitors Sysmon Process Creation events and identifies the execution of known LOLBins. For each event, the process name, command-line arguments, parent process, executing user, and host are captured to provide analysts with sufficient context for investigation.

SPL Query

```
index=main sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"
EventCode=1
(Image="*\powershell.exe" OR
Image="*\cmd.exe" OR
Image="*\rundll32.exe" OR
Image="*\regsvr32.exe" OR
Image="*\mshta.exe" OR
Image="*\certutil.exe" OR
Image="*\wmic.exe" OR
Image="*\bitsadmin.exe")
| table _time Computer User ParentImage Image CommandLine
```

Investigation Steps

1. Identify the LOLBin that was executed.
2. Review the parent process responsible for launching the binary.
3. Examine the complete command-line arguments.
4. Verify whether the execution aligns with expected administrative activity.
5. Correlate the event with additional endpoint telemetry such as network connections, registry changes, or PowerShell execution.

Expected Result

The detection identifies executions of commonly abused Windows native binaries and provides analysts with the context required to distinguish legitimate administrative usage from potentially malicious activity.

Analyst Notes

LOLBins are extensively used by Windows and enterprise applications during normal operations. The presence of a LOLBin alone does not indicate malicious activity. Investigation should focus on execution context, command-line parameters, parent-child process relationships, user account, and correlation with other suspicious events before determining whether an incident has occurred.

Key Takeaways

- LOLBins are commonly abused during real-world attacks.
- Behavioral context is more valuable than process name alone.
- Parent-child process analysis significantly improves detection accuracy.
- Monitoring LOLBins strengthens detection coverage across multiple MITRE ATT&CK tactics.

Security Dashboard Design

Purpose

The Security Dashboard provides a centralized view of endpoint activity, authentication events, detection results, and security metrics. It enables analysts to monitor the environment, identify anomalies, and prioritize investigations from a single interface.

Dashboard Objectives

The dashboard was designed to:

- Monitor endpoint security events in near real time.
- Visualize detection trends and authentication activity.
- Highlight suspicious behaviors identified by detection rules.
- Improve analyst visibility during security investigations.
- Support faster incident triage and decision-making.

Dashboard Components

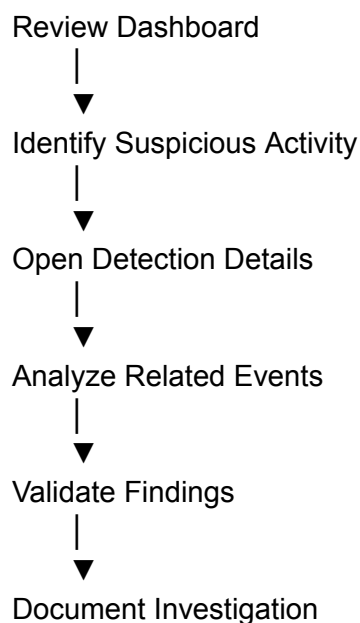
Dashboard Panel	Description
Total Security Events	Displays the total number of indexed security events.
Process Creation Activity	Shows process creation trends collected from Sysmon Event ID 1.
Failed Logons	Visualizes Windows authentication failures (Event ID 4625).

Account Lockouts	Displays Windows account lockout events (Event ID 4740).
PowerShell Activity	Monitors PowerShell process execution.
Network Connections	Displays outbound network connection events captured by Sysmon.
Detection Summary	Shows detections triggered by custom SPL rules.
Top Hosts	Identifies systems generating the highest number of security events.

Dashboard Benefits

- Centralized operational visibility.
- Faster identification of abnormal activity.
- Improved detection monitoring.
- Simplified investigation workflow.
- Better understanding of endpoint behavior.
- Reduced analyst response time.

Analyst Workflow



Key Takeaways

- Dashboards improve analyst efficiency by consolidating security information.
- Visualizing endpoint activity helps identify trends and anomalies.
- Detection summaries support rapid incident triage.
- Centralized monitoring strengthens day-to-day SOC operations.

Alert Configuration

Purpose

Alerting enables the Security Operations Center (SOC) to receive timely notifications when predefined detection rules identify suspicious or potentially malicious activities. The objective is to reduce detection time and support rapid investigation of security events.

Alerting Strategy

Alerts are generated from validated SPL detection rules and are designed to notify analysts only when predefined conditions are met. This approach minimizes false positives while ensuring that high-value security events receive immediate attention.

Configured Security Alerts

Alert Name	Trigger Condition	P r i o r i t y
Suspicious PowerShell Execution	PowerShell process detected	H i g h
Encoded PowerShell Command	Encoded command identified	H i g h
LSASS Process Access	Access to LSASS process detected	C r i t

		i c a l
Registry Persistence	Registry Run/RunOnce modified	H i g h
Failed Logon Threshold	Multiple failed logons detected	M e d i u m
Account Lockout	Windows account lockout event	H i g h
LOLBin Execution	Execution of monitored LOLBins	H i g h

Alert Workflow

Detection Rule

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SPL Query Executes

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Trigger Condition Met

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Security Alert Generated

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SOC Analyst Review

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Investigation & Response

Alert Design Principles

- Generate alerts only for validated detections.
- Prioritize alerts based on security impact.
- Include sufficient event context for investigation.
- Reduce duplicate or repetitive alerts.
- Support rapid analyst triage.

Operational Benefits

- Reduces Mean Time to Detect (MTTD).
- Improves analyst response efficiency.
- Prioritizes high-risk security events.
- Supports consistent incident handling.
- Enables faster investigation and decision-making.

Key Takeaways

- Effective alerts provide actionable information rather than excessive notifications.
- Alert quality is more important than alert quantity.
- Well-tuned detection rules help minimize false positives.
- Structured alerting improves SOC operational efficiency and investigation speed.

Investigation Workflow

Purpose

The investigation workflow provides a structured process for analyzing security alerts, validating suspicious activity, determining the scope of an incident, and documenting findings. A consistent workflow helps analysts perform accurate and repeatable investigations.

Investigation Process

When a detection rule generates an alert, the analyst follows a standardized investigation process to determine whether the activity is legitimate or represents a security incident.

Investigation Workflow

Security Alert



Review Alert Details



Validate Detection



Analyze Related Events



Determine Impact



Document Findings



Close or Escalate Incident

Investigation Steps

Step	Activity
Alert Review	Examine the alert, timestamp, host, and affected user.
Event Validation	Verify that the detection accurately represents the observed activity.
Event Correlation	Review related Windows and Sysmon events for additional context.
Scope Assessment	Determine whether the activity affects a single endpoint or multiple systems.
Risk Assessment	Evaluate the potential business and security impact.
Documentation	Record investigation findings, evidence, and analyst observations.
Escalation	Escalate confirmed security incidents according to the incident response process.

Investigation Checklist

- Identify the affected endpoint.
- Identify the associated user account.
- Review the initiating process.
- Examine command-line arguments (if applicable).
- Correlate related security events.
- Validate whether the activity is expected or suspicious.
- Record investigation results.

Investigation Outcomes

Outcome	Description
False Positive	Legitimate activity requiring no further action.
Suspicious Activity	Additional monitoring or validation required.
Confirmed Security Incident	Malicious activity confirmed and escalated for incident response.

Key Takeaways

- A structured investigation process improves consistency across security operations.
- Event correlation provides valuable context during incident analysis.
- Accurate documentation supports knowledge sharing and future investigations.
- Consistent triage and validation help reduce investigation time and improve response quality.

MITRE ATT&CK Coverage Matrix

Purpose

The MITRE ATT&CK framework was used to map detection rules to known adversary tactics and techniques. This alignment provides a standardized approach for evaluating detection coverage and identifying monitoring gaps within the laboratory.

Coverage Overview

The implemented detections focus on common attacker behaviors observed during endpoint compromise, including execution, credential access, persistence, discovery, defense evasion, and command-and-control activities.

Detection Coverage Matrix

Detection	ATT&CK Tactic	ATT&CK Technique	Technique ID
Suspicious PowerShell Execution	Execution	PowerShell	T1059.001
Encoded PowerShell Commands	Defense Evasion / Execution	PowerShell	T1059.001
LSASS Memory Dump	Credential Access	OS Credential Dumping	T1003.001
Registry Persistence	Persistence	Registry Run Keys / Startup Folder	T1547.001
System Discovery Commands	Discovery	System Information Discovery	T1082
Privileged Logon	Privilege Escalation	Valid Accounts	T1078

Failed Logon Detection	Credential Access	Brute Force	T1110
Suspicious Network Connections	Command and Control	Application Layer Protocol	T1071
Account Lockout Detection	Credential Access	Brute Force	T1110
LOLBin Execution	Defense Evasion	System Binary Proxy Execution	T1218

Coverage Summary

ATT&CK Tactic	Coverage
Execution	✓
Persistence	✓
Privilege Escalation	✓
Defense Evasion	✓
Credential Access	✓
Discovery	✓
Command and Control	✓

Benefits

- Standardizes detection documentation.
- Improves understanding of attacker behavior.
- Identifies detection coverage across the attack lifecycle.
- Simplifies communication between SOC analysts and security teams.
- Supports future expansion of detection capabilities.

Key Takeaways

- MITRE ATT&CK provides a common language for detection engineering.
- Mapping detections to ATT&CK improves visibility into monitoring coverage.
- The current implementation covers multiple high-value attacker tactics commonly observed in enterprise environments.
- Additional detections can be added over time to further expand ATT&CK coverage.

Detection Validation

Purpose

Detection validation verifies that each implemented detection rule accurately identifies the intended security event while minimizing false positives. This process ensures that detections are reliable, repeatable, and operationally effective.

Validation Approach

Each detection was tested by generating the corresponding Windows or Sysmon event within the laboratory environment. The generated telemetry was forwarded to Splunk Enterprise, where the associated SPL query was executed to confirm successful detection.

Validation Workflow

Simulate Activity

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Generate Windows / Sysmon Event

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Forward Event to Splunk

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Execute SPL Query

|



Verify Detection Results

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Tune Detection (If Required)

Validation Summary

Detection	Validation Status
Suspicious PowerShell Execution	✓ Validated
Encoded PowerShell Commands	✓ Validated
LSASS Memory Dump Detection	✓ Validated
Registry Persistence Detection	✓ Validated
System Discovery Detection	✓ Validated
Privileged Logon Detection	✓ Validated

Failed Logon Detection ✓ Validated

Suspicious Network Connection Detection ✓ Validated

Account Lockout Detection ✓ Validated

LOLBin Execution Detection ✓ Validated

Validation Criteria

The following criteria were used to evaluate each detection:

- Correct event was generated.
- Telemetry was successfully forwarded to Splunk.
- SPL query returned the expected results.
- Required event fields were available for investigation.
- Detection produced meaningful security context.
- No significant false positives were observed during testing.

Operational Benefits

- Confirms the accuracy of implemented detections.
- Verifies end-to-end telemetry collection.
- Improves analyst confidence in detection results.
- Reduces the likelihood of unreliable alerts.
- Supports continuous improvement of detection logic.

Key Takeaways

- Detection engineering is not complete until detections are validated.
- Reliable telemetry is essential for successful detection.
- Validation helps identify configuration issues before deployment.
- Continuous testing improves the quality and effectiveness of security monitoring.

Lessons Learned

Overview

Developing the Enterprise Detection Engineering & Threat Hunting Lab provided practical experience in building, validating, and documenting a security monitoring solution using enterprise methodologies. The project reinforced both technical and analytical skills required in a modern Security Operations Center (SOC).

Key Learnings

Endpoint Telemetry is Critical

High-quality endpoint telemetry is the foundation of effective detection engineering. Sysmon provides significantly greater visibility than native Windows logging, enabling more accurate detections and investigations.

Detection Requires Context

A single security event rarely confirms malicious activity. Effective detections combine process information, command-line arguments, user context, parent-child relationships, and related events to improve confidence and reduce false positives.

Detection Rules Must Be Validated

Writing an SPL query is only one part of detection engineering. Every detection should be tested with representative activity to confirm that it produces accurate and reliable results.

Threat Hunting Complements Detection

Not every threat generates an alert. Proactive threat hunting helps identify suspicious behaviors that may bypass existing detection rules and contributes to continuous improvement of detection coverage.

Documentation Matters

Well-structured technical documentation improves project maintainability, supports knowledge sharing, and enables other analysts to understand and reproduce the implementation.

Skills Strengthened

- Detection Engineering
- Splunk Search Processing Language (SPL)
- Windows Event Analysis

- Sysmon Telemetry Analysis
- Threat Hunting
- Security Investigation
- MITRE ATT&CK Mapping
- Technical Documentation

Operational Insights

- Accurate telemetry improves investigation quality.
- Well-designed detections reduce false positives.
- Standardized investigation workflows improve analyst consistency.
- Continuous tuning is essential for maintaining effective security monitoring.

Key Takeaways

- Effective security monitoring depends on reliable telemetry, validated detections, and structured investigations.
- Detection engineering is an iterative process that evolves as new threats emerge.
- Practical implementation provides a deeper understanding of SOC operations than theoretical study alone.

Future Enhancements

Overview

The current implementation provides a strong foundation for detection engineering and threat hunting. As the laboratory evolves, additional capabilities can be integrated to improve detection coverage, automation, and operational efficiency.

Planned Enhancements

Active Directory Integration

Expand the monitoring environment by integrating Active Directory domain controllers to detect authentication anomalies, privilege changes, group modifications, and account management events.

Microsoft Sentinel Integration

Extend the laboratory into a hybrid SIEM environment by forwarding security telemetry to Microsoft Sentinel. This will enable cloud-native detection engineering, KQL-based threat hunting, and analytics rule development.

Sysmon Configuration Optimization

Enhance the Sysmon configuration by refining event filtering, improving telemetry quality, and reducing unnecessary log volume while maintaining strong detection coverage.

Expanded Detection Library

Develop additional detection rules covering:

- Lateral Movement
- Scheduled Task Persistence
- Remote Service Creation
- WMI Activity
- RDP Monitoring
- Service Installation
- DLL Side-Loading
- Credential Abuse

Detection Automation

Implement automated alerting and scheduled searches to improve detection efficiency and reduce manual monitoring efforts.

Threat Hunting Expansion

Develop additional threat hunting playbooks focused on persistence, privilege escalation, lateral movement, and command-and-control techniques.

Dashboard Enhancements

Expand the dashboard with additional operational metrics, ATT&CK coverage visualization, detection trends, and analyst performance indicators.

Long-Term Goals

- Increase detection coverage across additional ATT&CK tactics.
- Improve detection accuracy through continuous tuning.
- Expand telemetry sources beyond a single endpoint.
- Build a more comprehensive enterprise SOC simulation.
- Continue developing practical detection engineering skills.

Key Takeaways

- Security monitoring is a continuous improvement process.
- Detection engineering evolves as attacker techniques change.
- Future enhancements will improve visibility, scalability, and detection coverage while maintaining the core principles established in this implementation.

Conclusion

The **Enterprise Detection Engineering & Threat Hunting Lab** was successfully designed and implemented to simulate the core functions of a modern Security Operations Center (SOC). The project demonstrates how endpoint telemetry can be transformed into actionable security intelligence through centralized log collection, behavioral detection engineering, proactive threat hunting, and structured security investigations.

By integrating Microsoft Windows, Sysmon, Splunk Universal Forwarder, and Splunk Enterprise, the laboratory establishes a complete monitoring pipeline that supports the collection, analysis, and investigation of security events. The implementation validates the importance of high-quality telemetry, reliable log management, and well-designed detection logic in identifying suspicious activities within an enterprise environment.

Throughout the project, custom SPL detection rules were developed and validated to monitor multiple MITRE ATT&CK tactics, including Execution, Persistence, Credential Access, Discovery, Defense Evasion, Privilege Escalation, and Command and Control. Interactive dashboards, structured investigation workflows, and documented detection methodologies further demonstrate how security analysts can efficiently monitor, analyze, and respond to potential threats.

Beyond the technical implementation, the project emphasizes industry best practices such as standardized documentation, detection validation, continuous improvement, and behavioral analysis. These practices are essential for building reliable security monitoring capabilities and reflect the operational approach followed by enterprise Security Operations Centers.

This laboratory serves as both a practical learning platform and a professional demonstration of detection engineering skills. It highlights the ability to design monitoring solutions, develop behavioral detections, analyze endpoint telemetry, investigate security events, and document technical implementations using structured engineering practices.

As cybersecurity threats continue to evolve, effective detection engineering and proactive threat hunting remain fundamental to modern defensive strategies. The knowledge and practical experience gained through this project provide a strong foundation for contributing to SOC operations, incident detection, and continuous security monitoring within enterprise environments.